

Graphics Unit

Enniskillen Royal Grammar| Year 9

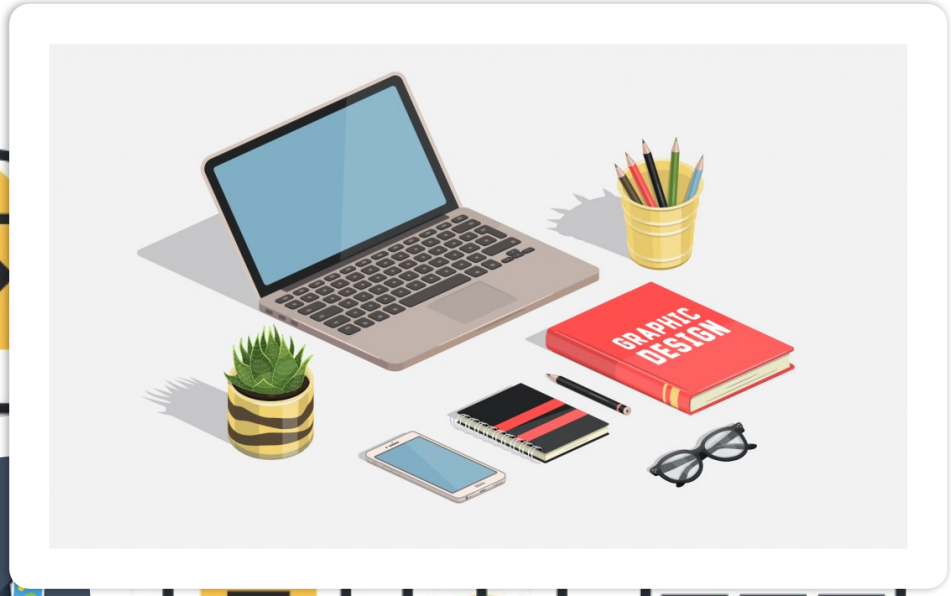
This is a **BORING** paragraph, trying to describe what **GRAPHICS** are all about.

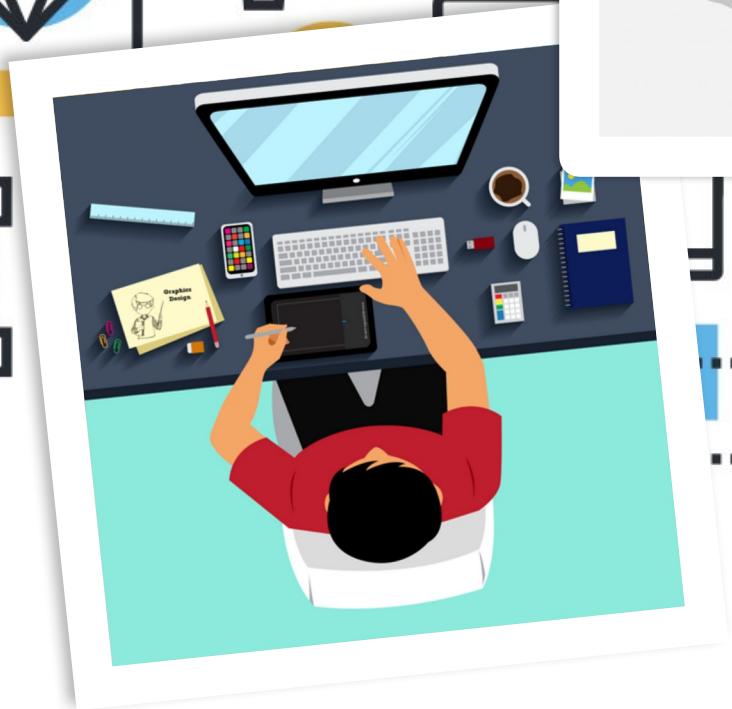
Not very interesting is it!

This is what the internet was like in the beginning – nothing but text… and more text… and even more text…

Now look what a difference graphics can make...







Much more interesting! Graphics bring colour and interest all around us.
This lesson is all about why and how we use graphics.

How Graphics are Used

- Graphics are seen everywhere around us. Images are used to convey all kinds of information and messages.

- Here are some common forms of use:



Logos

Companies are keen to make sure you recognise their product from a distance, and so a simple image is used that lets you know immediately who they are.



Logos

They are so valuable to a company. They are protected by law to prevent anyone else from using them. This protection is called a 'trademark'

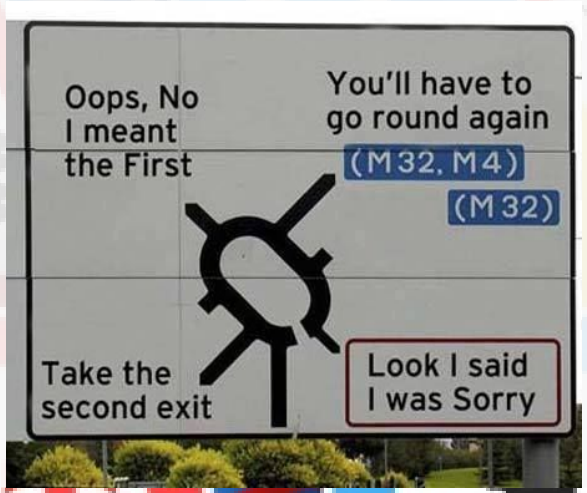


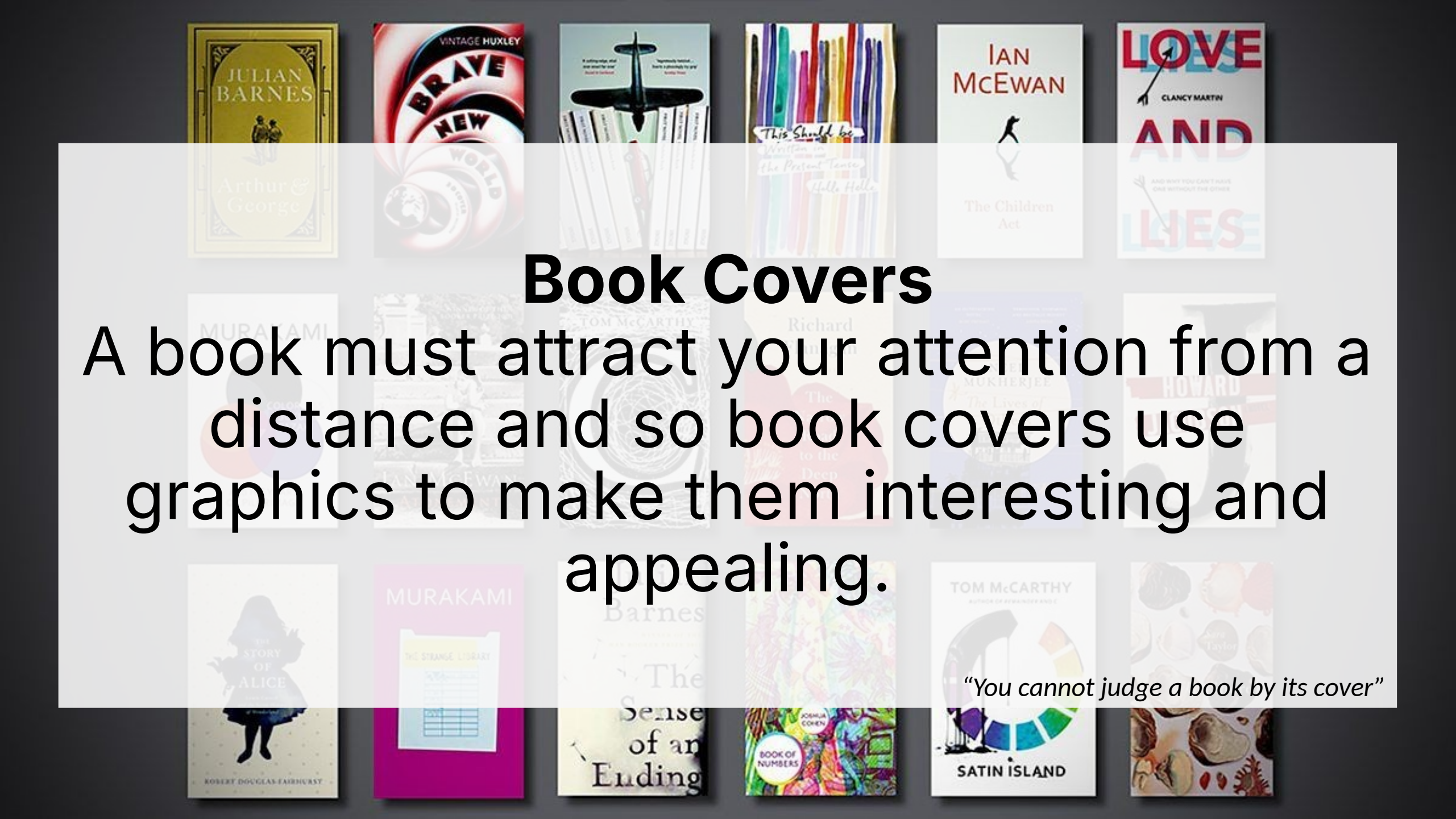
Road Signs

Signs provide information using graphics, often without words. They are excellent in places such as airports, city streets, roads and so on.

Road Signs

Signs need to be instantly recognisable and be able to overcome any language barriers.





Book Covers

A book must attract your attention from a distance and so book covers use graphics to make them interesting and appealing.

"You cannot judge a book by its cover"



Magazines

There are so many magazines on the market that the companies have their work cut out to persuade you to buy their product. They make good use of graphics to make their covers eye-catching and vibrant. Strong images are missed with text.

Product Packaging

The way a product is presented can have a great effect on what you think of it. Manufacturers spend millions of pounds developing the packaging to make it attractive to you. Most packaging has some form of graphic on it. Next time you buy something have a look.

Activity



- Mr Butler is looking for a new logo for his Computer Graphics Company.

- He wants to have:
  - 3-5 colours used
  - A simple image or text
  - Clear, easily printed logo.

- Use an A4 page to design a logo which is eye catching, clear and tells potential customers what the company can offer!
- Upload a photo of this to google classroom for marking!



Computer Generated Imagery (CGI)

- Computer generated imagery means that computer graphics are used to create special effects in movies and computer/ video games.

- Specialist computer software is used to make the CGI for movies but it requires a huge amount of skill in order to create something which looks realistic.



Computer Generated Imagery (CGI)

- More powerful computers have allowed CGI to be used in many of the movies made today for example, Lord of the Rings, Pirates of the Carribean and Avengers where the CGI is mixed with human actors.

- Whole movies can also be made with just CGI, such as Shrek, Ice Age and Polar Express.

- CGI is now so good it can be hard to tell which is real and what is made by the computer.



Turn off your monitor.

- You will need to work with the person next to you for the next two minutes.

- You will only need a scrap piece of paper and a pen

<u>The Question:</u>

List all of the different SOURCES of graphics you can.



Sources of Graphics

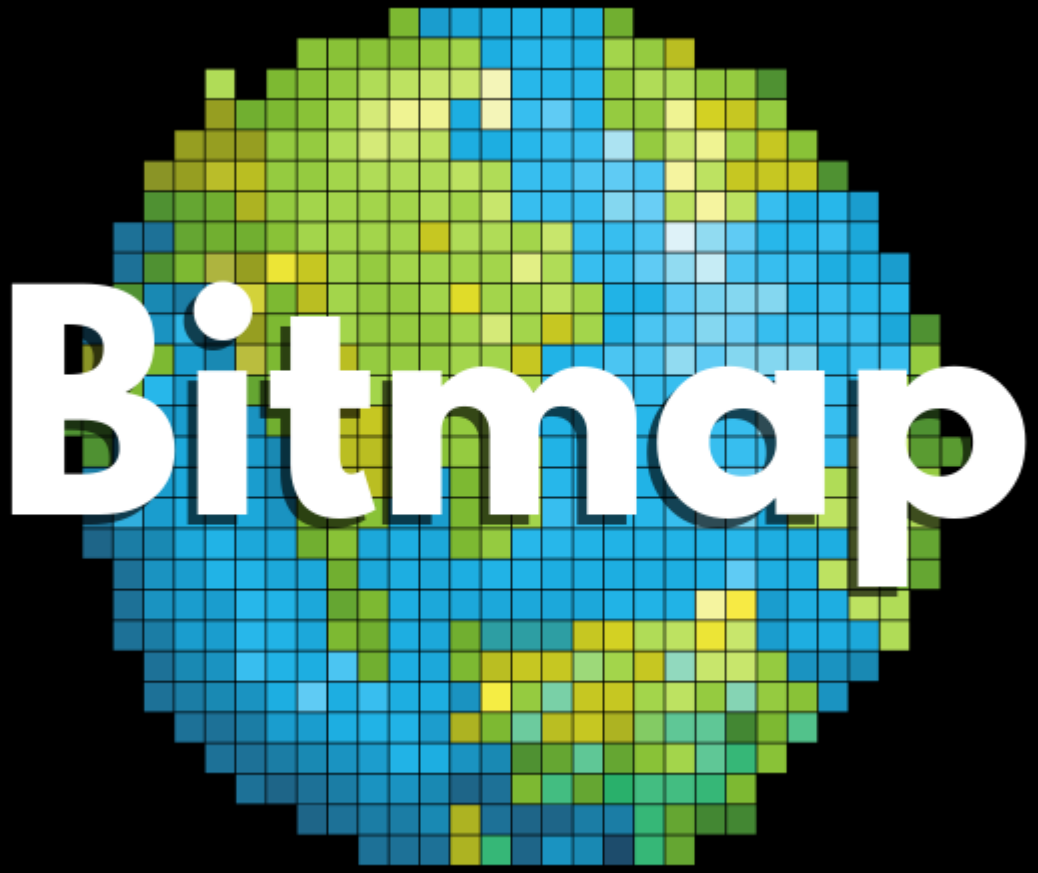
Graphics can be obtained from various places. The main ones are:

- Clipart
- Internet (Google Images/Flikr/other photo sharing sites)
- CDs/DVDs
- Digital Camera
- Hand Drawn Images (scanned in)
- Magazines (scanned in)

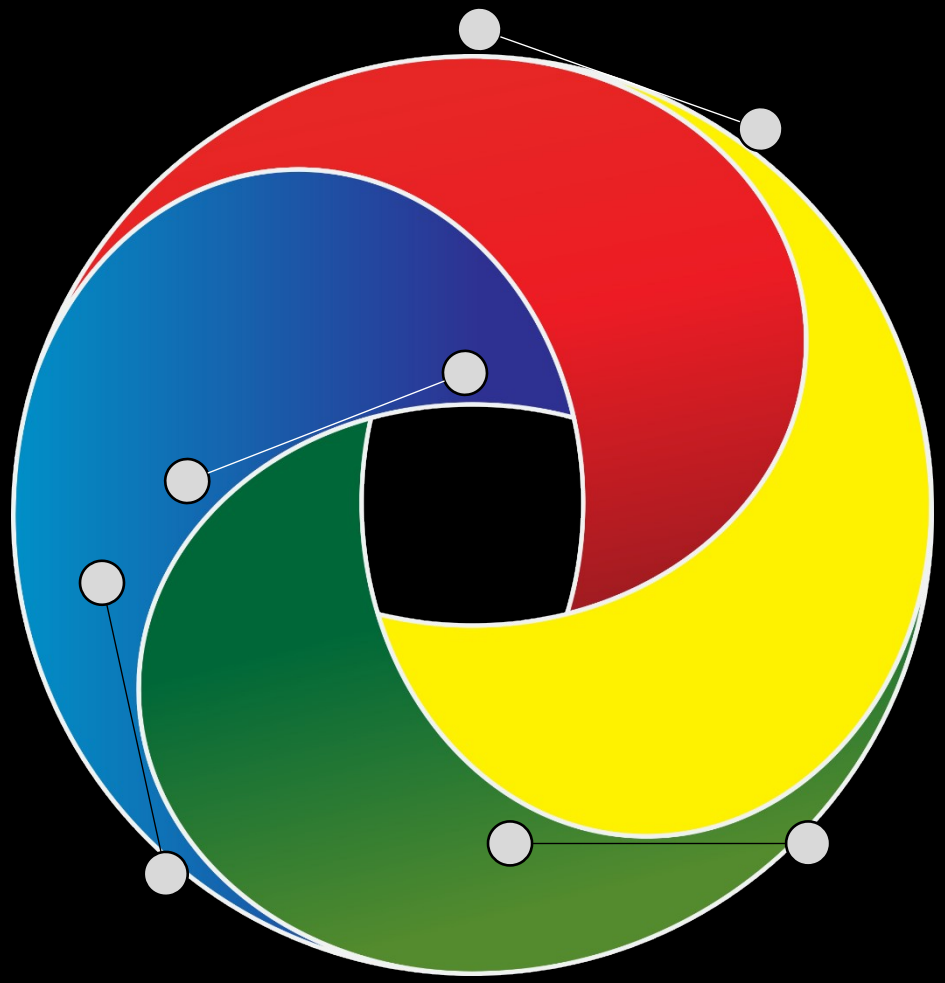
NOTE: A scanner is NOT a source of graphics, it is a device for importing graphics from paper onto the computer.

Bitmap Images

- Bitmap images are made up of a grid of tiny square dots called **pixels**.
- Each pixel can be its **own colour** – but **only one colour** at any one time!
- When you zoom into a bitmap image, it is possible to see these individual pixels, or you see these ‘jagged’ edges. This is called **pixilation**.
- Bitmaps store information on each individual pixel, meaning it is perfect for highly detailed photographs – however, can result in large file sizes.



Vector Images



- Some graphics do not use a grid of pixels, but instead are ‘calculated’ using mathematical equations to create lines, curves and shapes.
- It’s easily scalable/resizable as it doesn’t rely on fixed point references or grids to create its shapes.
- Although, think of how many individual shapes would be required for a photo – too many individual shapes so Vector graphics should only be used for simple images such as logos.



- Design your own MINECRAFT, EMOJI or MARIO pixel image in MS Excel or MS Word using the table facility (10 x 10) to design a 2-bit colour image. You can create your own colour code.
- Below your grid, show how the image can be encoded as a grid of numbers.
- Use a Key to label your colour system:

00 = White 01 = Black

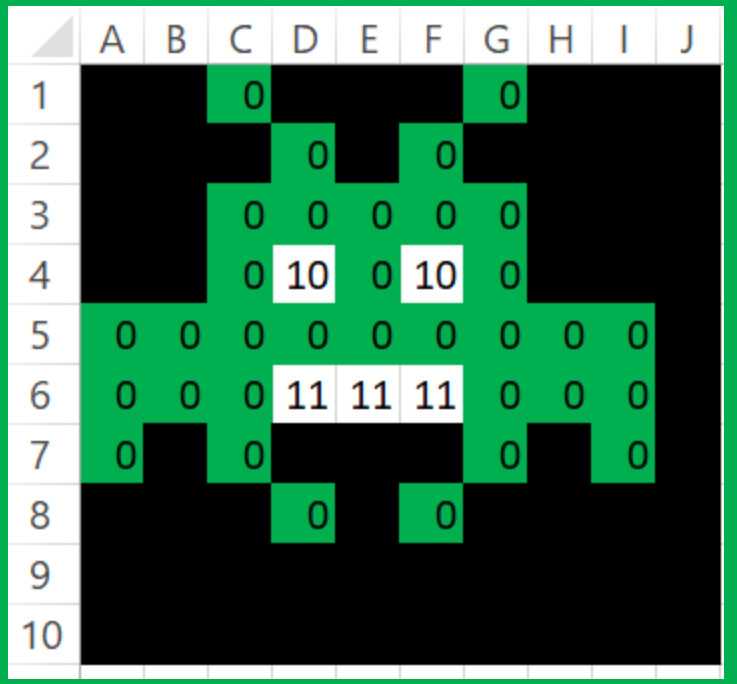
10 = 11 =

00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00

Now you have designed your Emoji,

1. Remove the background colouring of your image.
2. Place your ‘code’ in a single line
3. Send this ‘code’ to your partner (the person next to you)
4. At the end of the email, also give them the ‘key’.

5. Once you receive this ‘file’, recreate your partners image on your screen using only the code sent (Don’t just copy what is on their screen!)



Bitmap and Vectors – What do you already know?

ACTIVITY

- **TASK:** In pairs, create two fact files; one of which will be for a **bmp** file and one for a **SVG** file.

- **Once complete, print** two **copies of each fact file.** Each person in the pair should keep a copy. Alternatively Email this to one another!













Computer Graphic Programs

- One of the most famous graphics packages is Adobe Photoshop, although there are numerous others which you can use, designed for a whole range of users. Examples include:

MS Paint

Adobe Illustrator

CorelDRAW

GIMP

PIXLR

Paintshop Pro

Picasa











- With these packages, there are several features that are common throughout. As PhotoShop is the most commonly used package, we will use it. However, GIMP and PIXLR are also highly recommended as they are free to use!

MS Paint

Adobe Illustrator

CorelDRAW

GIMP

PIXLR

Paintshop Pro

Picasa

Computer Graphic Programs











Computer Graphic Programs

- Although, some professional Comic Book Artists prefer to use Adobe Illustrator as it is an example of a Vector Based graphics package – so images can be scaled much easier. It's perfect for Logo Designers

MS Paint

Adobe Illustrator

CorelDRAW

GIMP

PIXLR

Paintshop Pro

Picasa

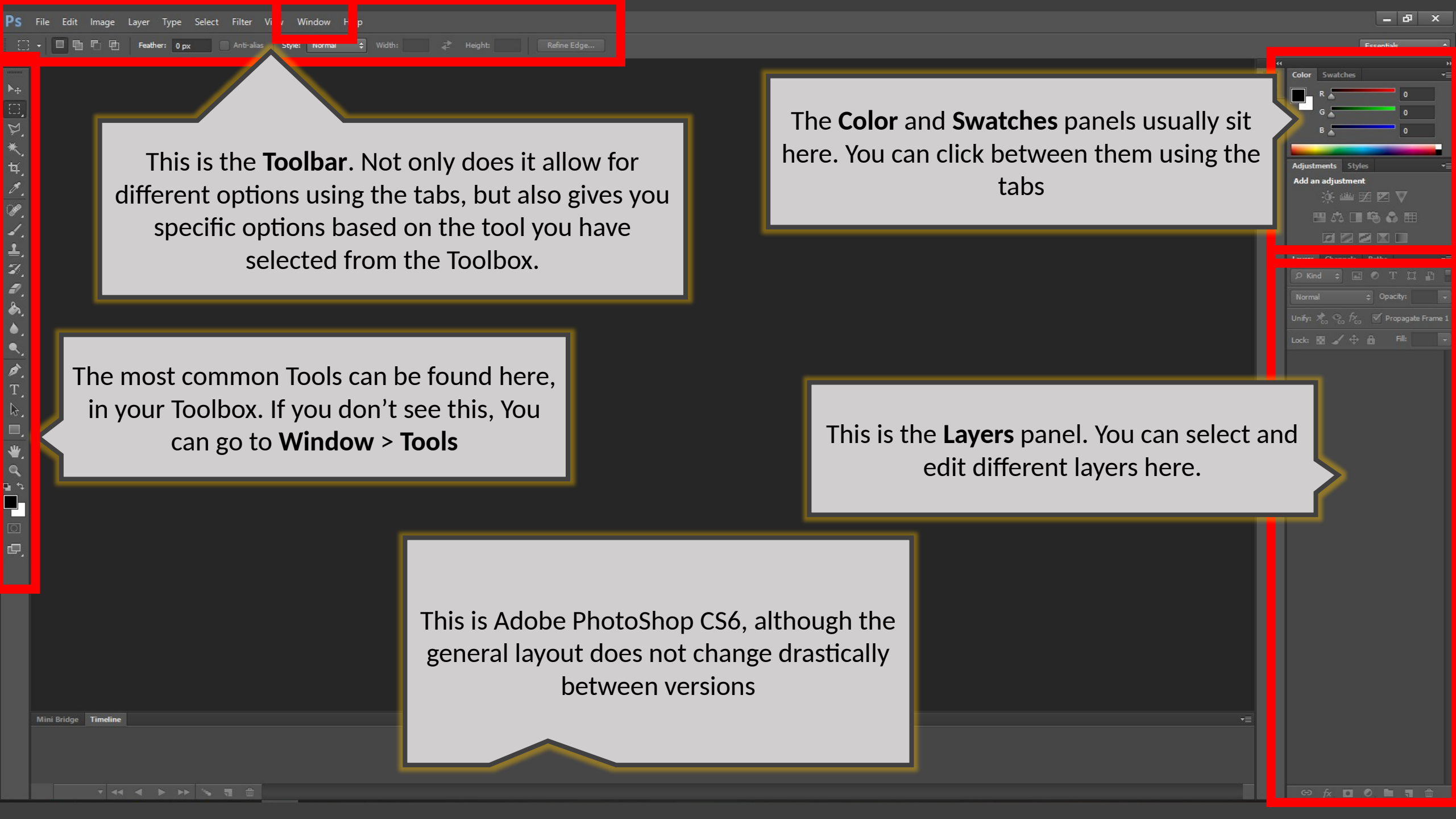
We will use Photoshop



- As we may wish to edit real photographs, which we have already said use Raster (bitmap) formats, we are going to use Adobe Photoshop for our Image Editing.

- We have this available in school, and can easily be imported and reopened in GIMP or PIXLR if you wanted to continue working at home.

So Let’s have a look at some of these features...



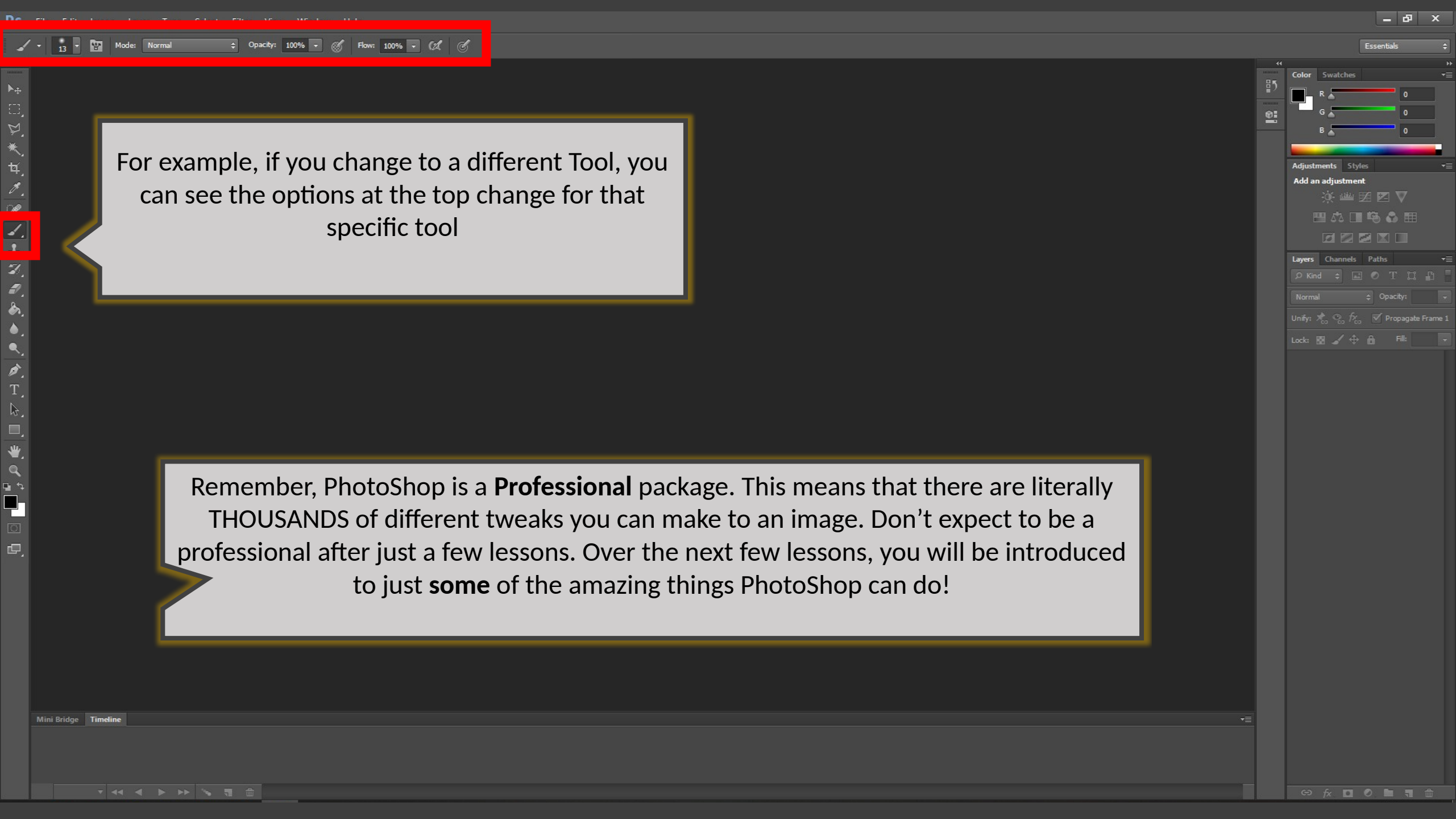
This is the **Toolbar**. Not only does it allow for different options using the tabs, but also gives you specific options based on the tool you have selected from the Toolbox.

The **Color** and **Swatches** panels usually sit here. You can click between them using the tabs

The most common Tools can be found here, in your Toolbox. If you don't see this, You can go to **Window** > **Tools**

This is the **Layers** panel. You can select and edit different layers here.

This is Adobe PhotoShop CS6, although the general layout does not change drastically between versions



For example, if you change to a different Tool, you can see the options at the top change for that specific tool

Remember, PhotoShop is a **Professional** package. This means that there are literally THOUSANDS of different tweaks you can make to an image. Don't expect to be a professional after just a few lessons. Over the next few lessons, you will be introduced to just **some** of the amazing things PhotoShop can do!

Let's have a look at the tool box first of all

This is the **Marquee** tool. It allows you to select part of an image. Holding Down the left mouse button lets you Change the shape of the marquee



Let's have a look at the tool box first of all



The **Move** tool lets you move the selection or whole layer around the screen.

Let's have a look at the tool box first of all

The **Lasso** tool is similar to the Marquee tool, however this allows you to select around a specific shape in an image. The Magnetic Lasso can be found by holding down the left click. This is used to 'stick' to a contrasting shape.



Let's have a look at the tool box first of all



This is the **Magic Wand** tool. Clicking on an area it will select pixels with the same or very similar colour. You can change the tolerance in the toolbar to allow a bigger or smaller range in the tool's selection tone.

Let's have a look at the tool box first of all

The **Crop** tool does the same as it does in Microsoft Word and other packages. It allows you to select an area of an image and discard everything outside this area.



Let's have a look at the tool box first of all



This is the **eye dropper** tool. It can get the colour from any part of the image.

Let's have a look at the tool box first of all

This less-used tool is the **Spot Healing** tool, as well as the **Patch** and even the **Red Eye** tool. These are designed to resolve blemishes and where photos capture red eyes if the flash is on.



Let's have a look at the tool box first of all

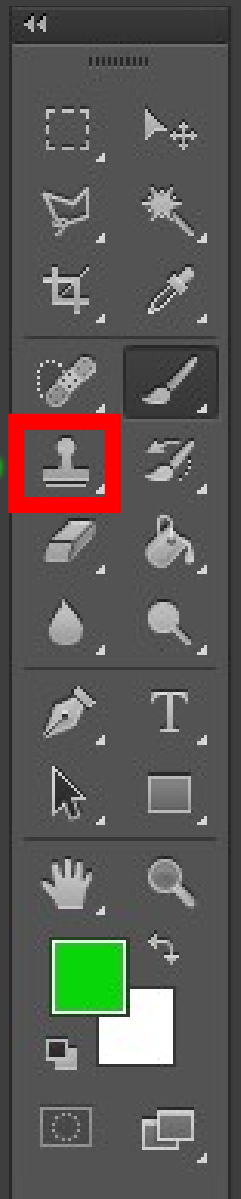


One of the most used tools is the **Brush** tool. It allows you to draw over your graphic using a variety of brush types. You can change the brush type, as well as size, from the toolbar.

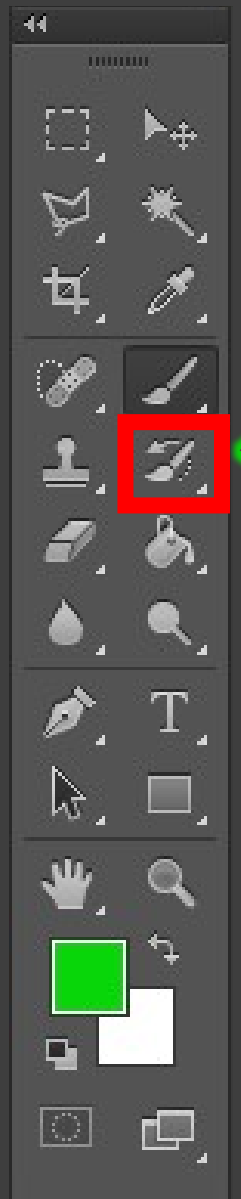
Let's have a look at the tool box first of all

This is the **Clone Stamp** tool. This tool is special as it allows you to hold **Left Mouse + Alt** on the keyboard to select an area on the screen.

When you release the click and click elsewhere, you can copy an area, in essence, 'clone' where the previous click was.



Let's have a look at the tool box first of all

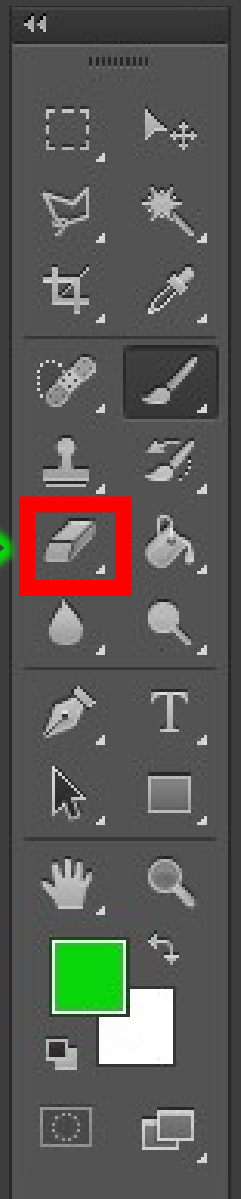


This **History Brush** tool can be used to 'undo' what has been done to a selected area of the screen.

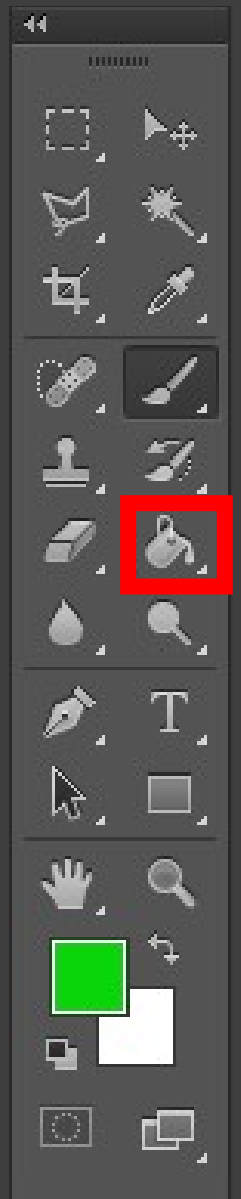
Use this tool to restore a portion of an image to an earlier state, while leaving the rest of the modified image alone

Let's have a look at the tool box first of all

The **Eraser** tool does exactly as it suggests – it rubs out part of the selected layer. It is just important you have selected the layer you wanted in the **Layers** panel at the bottom right of the screen!



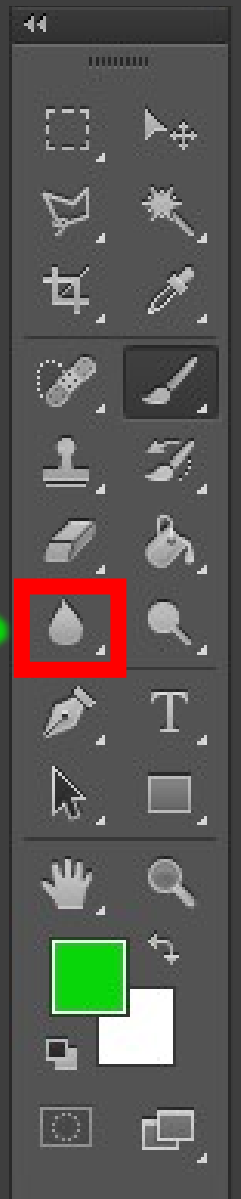
Let's have a look at the tool box first of all



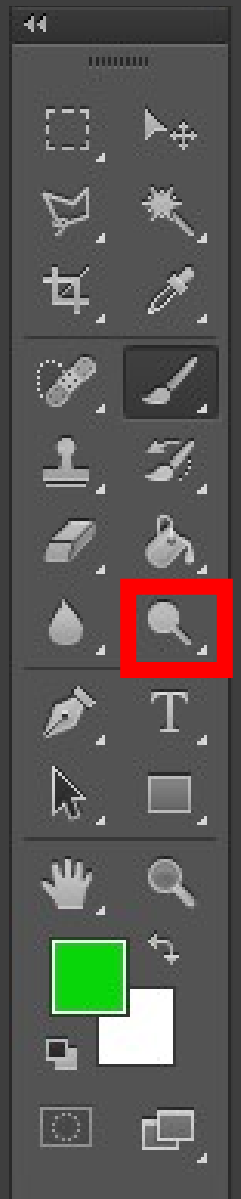
The **Paint Bucket** Tool allows an area to be filled with a solid colour. You can also find the **Gradient** tool here – allowing you to fill the selected area with a gradient instead!

Let's have a look at the tool box first of all

The **Blur**, **Sharpen**, and **Smudge** tools can be found here. They allow you to clarify the edges or even fade in edges of objects here.



Let's have a look at the tool box first of all



There are three tools here: the **Dodge**, **Burn** and **Sponge** tools can be found here. These let you lighten or darken areas – mainly used to make a 2D shape look more 3D by adding shadowing/shine effects.

Let's have a look at the tool box first of all

These next tools could realistically be placed together. The **Pen** and **Text** tools allow you to add freehand or font text onto the screen. It creates a separate layer on top of the currently selected layer.



Let's have a look at the tool box first of all



These tools are less commonly used, and allow for features such as editing the **Zoom** of the preview image or draw a basic rectangles/shapes on the screen.

Let's have a look at the tool box first of all



So this is the Colour selection. You can click on the Green and White squares to change the colours of the brush, or click the double arrow to switch these colours around. The icon in the bottom left allows you to reset the selected colours back to Black and White.

Well, now we know the theory...

LET’S GET SOME PRACTICAL!



Adobe Photoshop

STEP:	1
Tool/Skill Used:	Organisation
Purpose:	Ensure we can find all of our files and folders for future.

- Create a new folder called '**Simpsons Photoshop**' inside of your Year 9 > ICT Folder.
- Save the file **HOMER.png** into this folder.



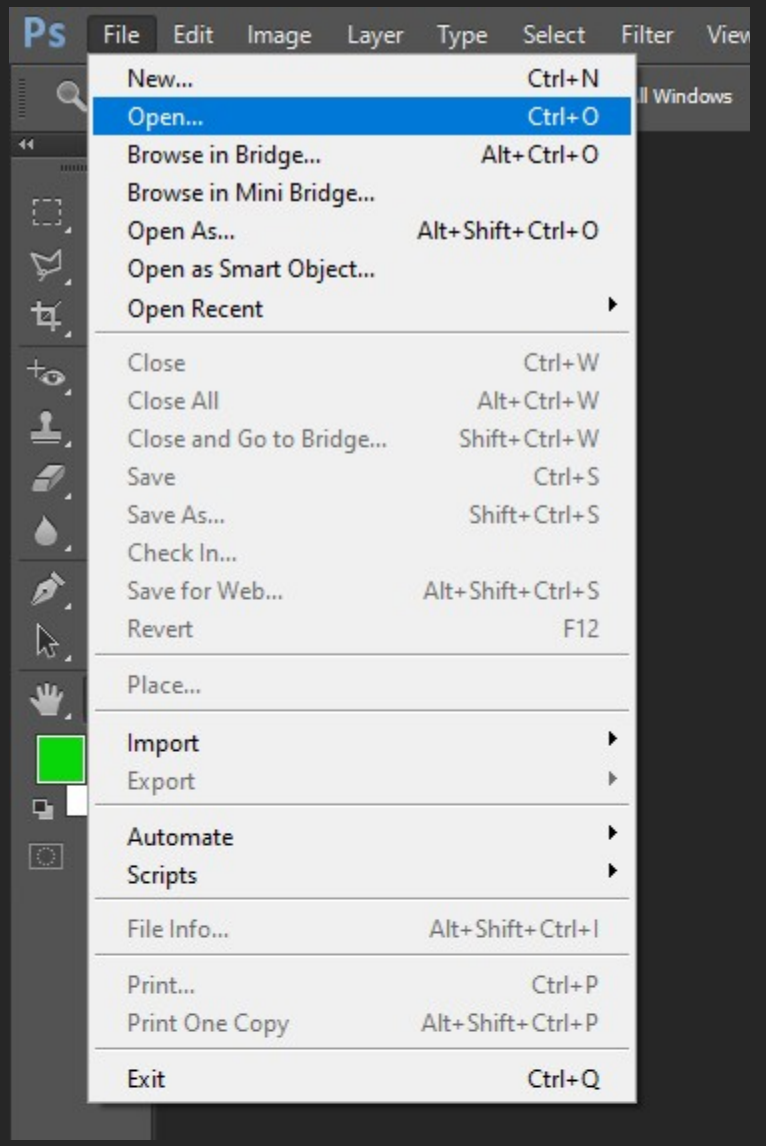
Adobe Photoshop

STEP: 2

Tool/Skill Used: Organisation

Purpose: Opening/Importing a graphic into Photoshop

- Open Adobe Photoshop from My Apps.
- Go to File > Open and open the HOMER.png image

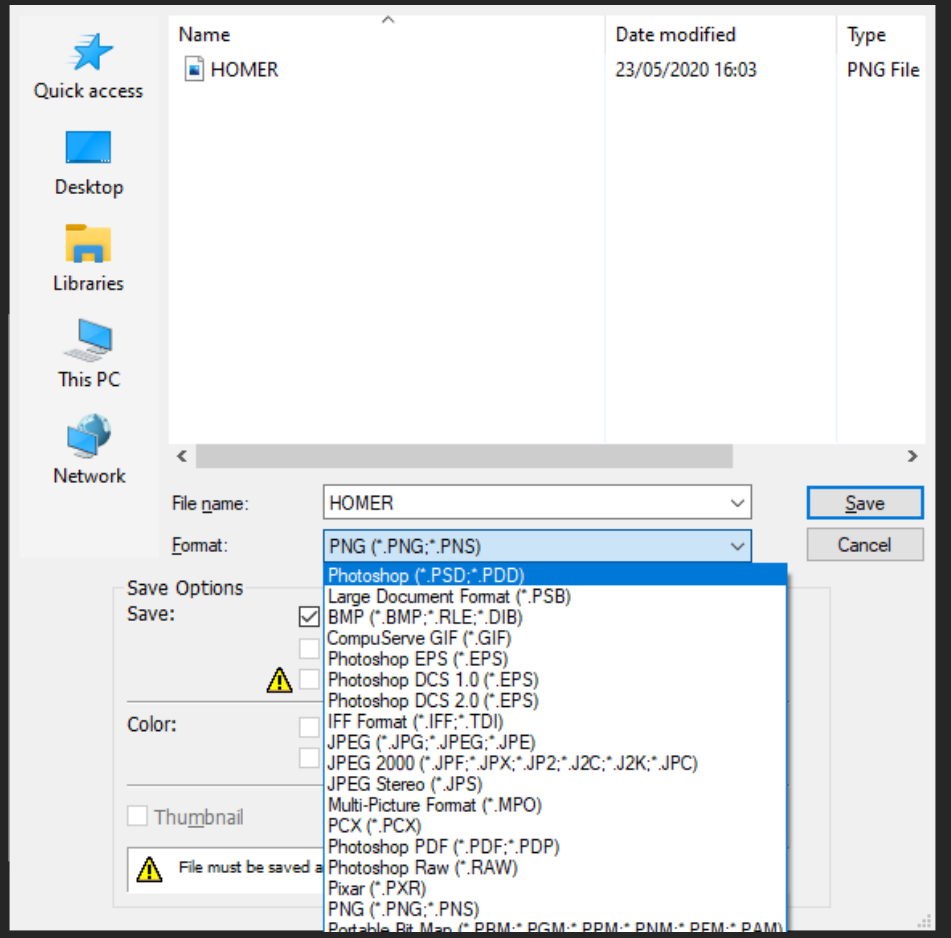


Adobe Photoshop

STEP: 3

Tool/Skill Used: Organisation

Purpose: Saving a Photoshop File



- Now We should have a Homer Simpson on screen.
- Before we do ANYTHING, go to **File > Save As...**
- We need to save this image as a **.psd** file

Challenge!



- You have **5 minutes** to ‘fix’ Homer to look like the image on the right.
- So,
  - No Background
  - No blue eyes
  - No green marks around hit mouth

a) YOU MUST USE AT LEAST 3 TOOLS FROM THE TOOLBOX

b) YOU CANNOT JUST COPY IN A ‘FIXED IMAGE’.

c) YOU CAN ONLY USE ONLY LAYER



Hands Up Discussion

Which Tool did you use to make his eyes white?

Which Tool did you use to remove the background?

Which Tool did you use to remove the green dots?

Remember there may be more than one right answer!

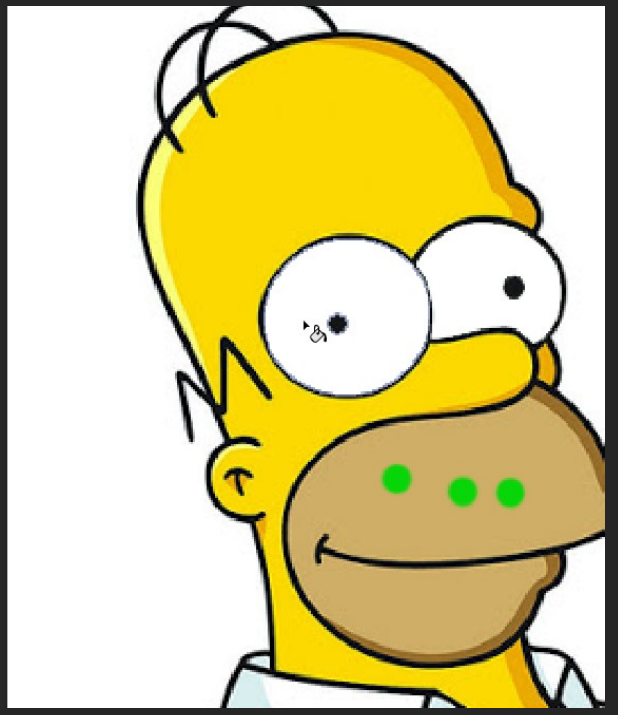
Adobe Photoshop

STEP: 4

Tool/Skill Used: Paint Bucket

Purpose: Make Eye Colour White





- Select the Paint Bucket Tool from the Toolbox
- Make sure the Colour White is selected
- Click on the blue eye to fill it white.

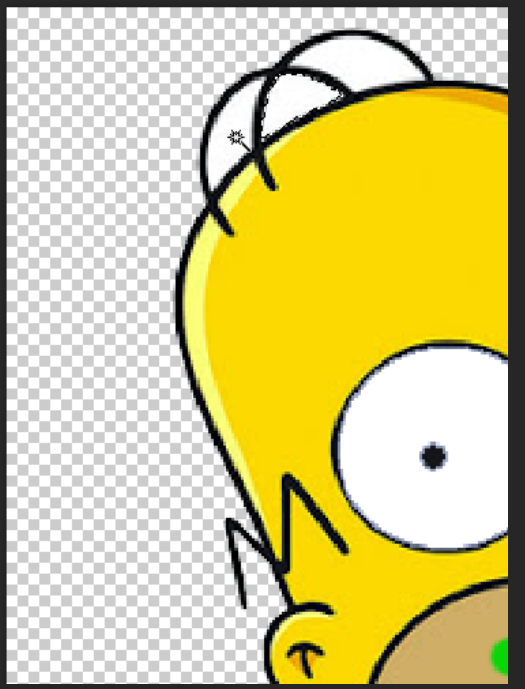
Adobe Photoshop

STEP: 5

Tool/Skill Used: Magic Wand

Purpose: Remove background colour





- Select the Magic Wand Tool from the Toolbox
- Click on the white background
- Press Delete (Del) on the keyboard
- Don’t forget to delete the white around his hair!

Adobe Photoshop

STEP: 6

Tool/Skill Used: Clone Stamp Tool

Purpose: Remove the green dots around Homer's beard.





- Select the Clone Stamp Tool
- Whilst Holding ALT, click on a 'good' section of Homer's beard, click.
- Release ALT.
- Now hover one of the green dots and click/drag to 'clone'.

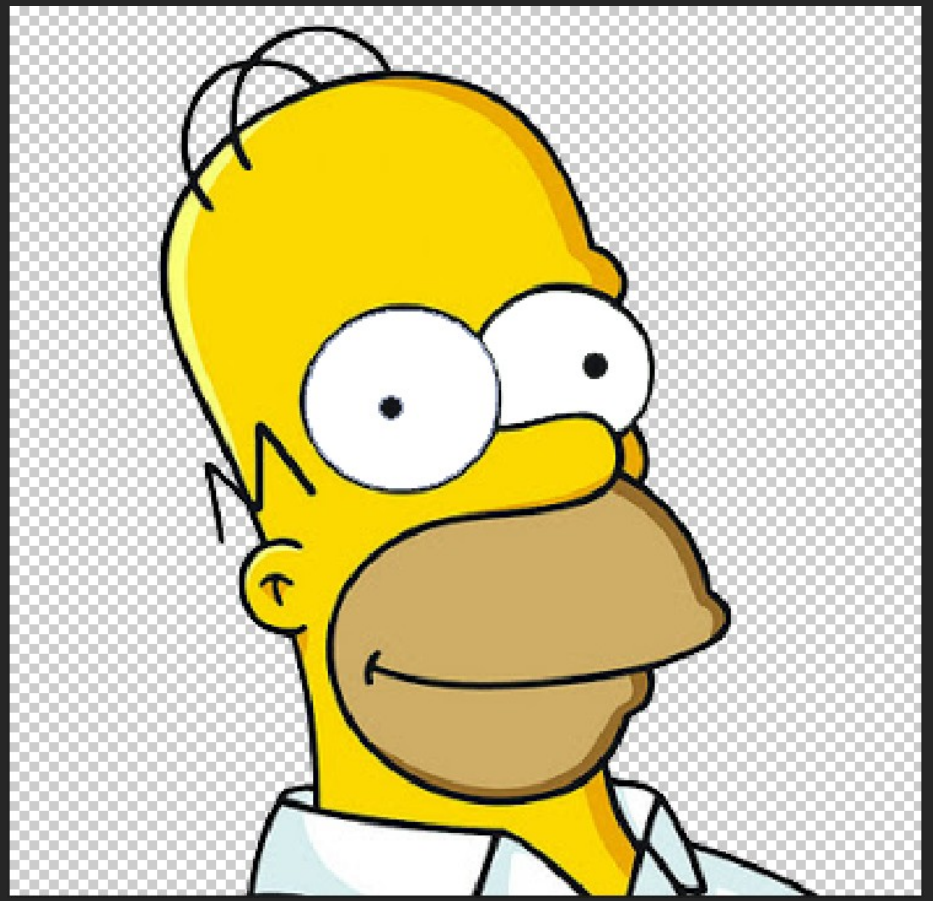
Adobe Photoshop

STEP: 7

Tool/Skill Used: Organising

Purpose: Save the PSD file and also as an image

- Go to **File > Save** to save the **.psd** (Photoshop) file.
- We also want to go to **File > Save As...** We need to select **.png** to save as an actual image!



Adobe Photoshop

STEP: 7

Tool/Skill Used: Organising

Purpose: Save the PSD file and also as an image

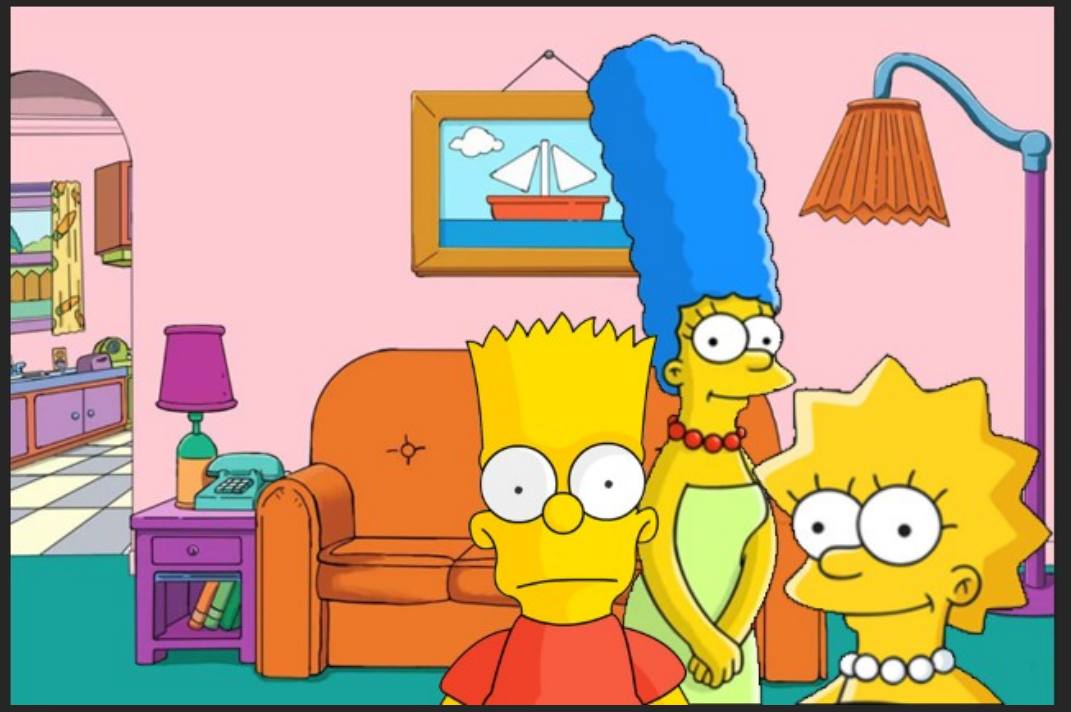


- Go to **File > Save** to save the **.psd** (Photoshop) file.
- We also want to go to **File > Save As…** We need to select **.png** to save as an actual image!

Adobe Photoshop

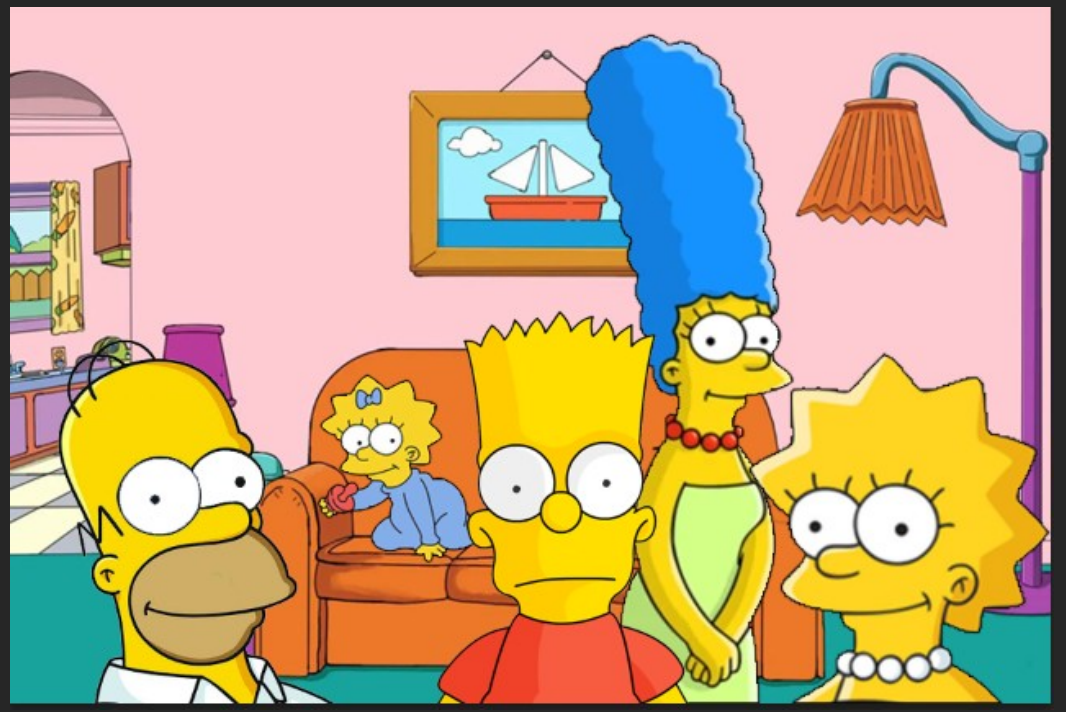
STEP:	8
Tool/Skill Used:	Layers
Purpose:	Experiment with Layers

- Open ‘Activity 2.png’ in Adobe Photoshop.
- Copy Homer.png and place him next to his family.
- Use the Cursor tool to move him into position



Adobe Photoshop

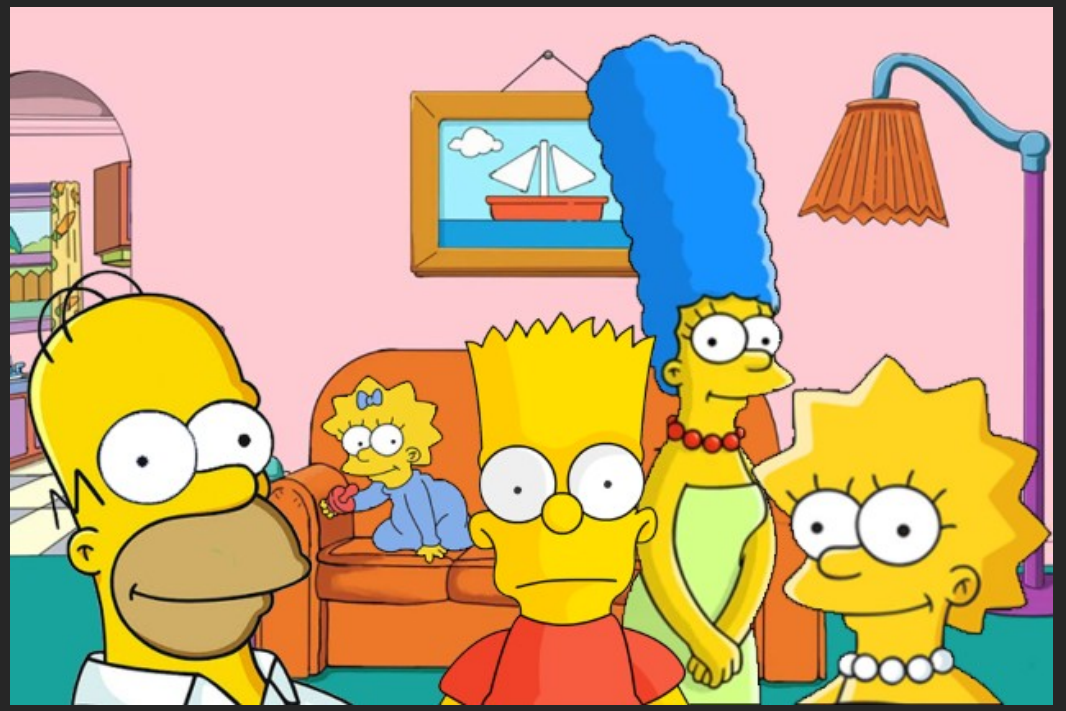
STEP:	9
Tool/Skill Used:	Layers
Purpose:	Experiment with Layers



- We have forgotten Maggie!
- Using the Skills you have learnt, find a picture of Maggie on the internet – add her into this family photo.
- Place her on the sofa looking at the camera.

Adobe Photoshop

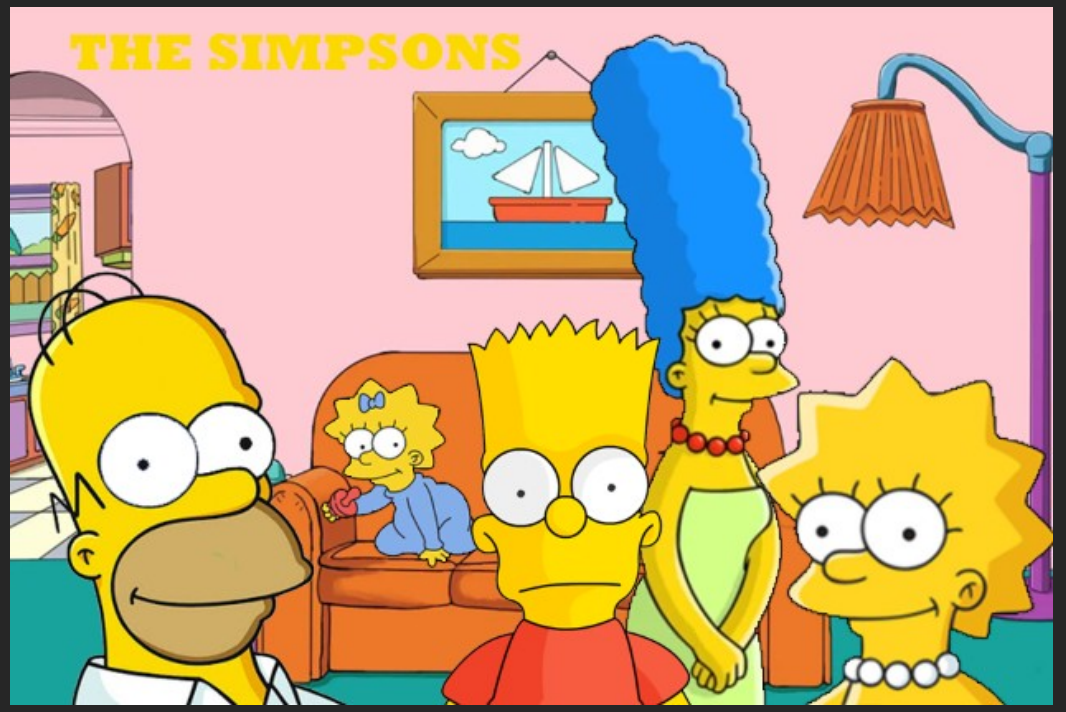
STEP:	10
Tool/Skill Used:	Sizing
Purpose:	Ensure layers are correctly sized in proportion to the rest of the image



- So, we have probably been able to add in these. However, did you find that Homer was a little… short? Let's fix that.
- Click on the Homer Layer
- Click on **Edit > Transform > Free Transform**. Or Press **Ctrl + T**. Hold **Shift** to maintain aspect ratio.
- Resize Homer to be a little more realistic in size.

Adobe Photoshop

STEP:	11
Tool/Skill Used:	Text/Font
Purpose:	Add text to your graphic.



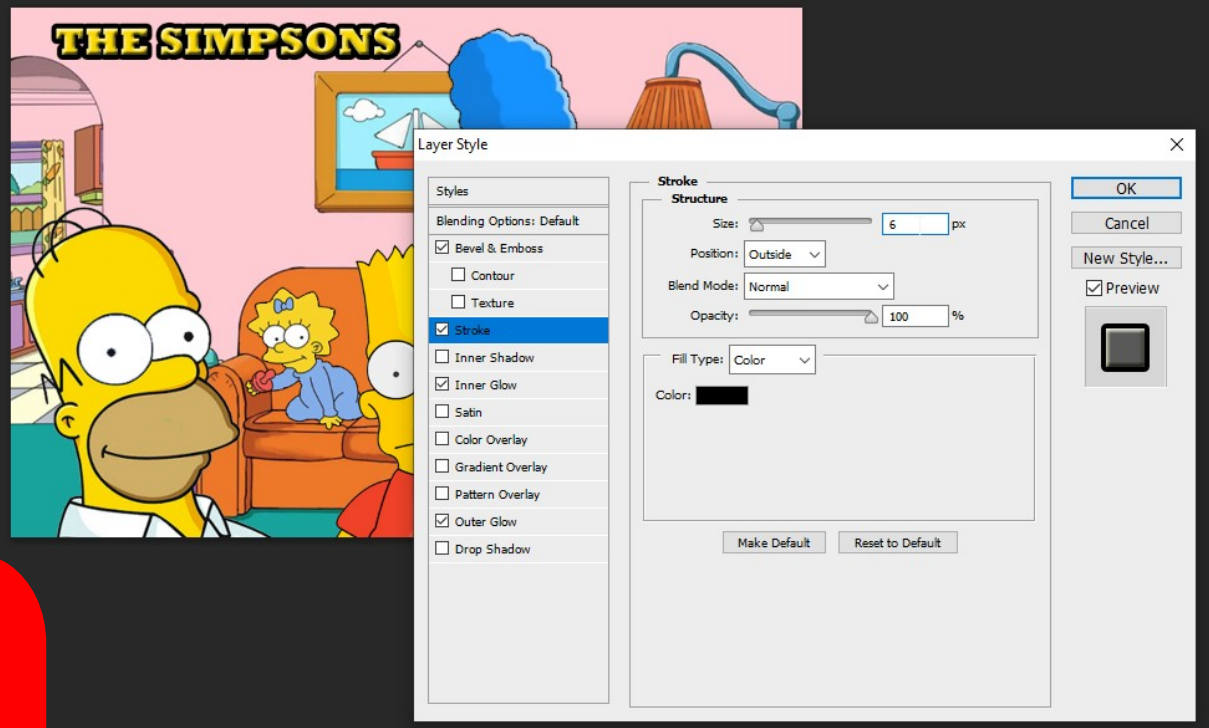
- Use the **Text** tool to add '**The Simpsons**' onto the screen.
- Position this into the **top left** of the screen.
- Make the Text **YELLOW**

Adobe Photoshop

STEP: 12

Tool/Skill Used: Blending Options

Purpose: Make the text on screen really stand out. Note: This can also be used on standard layers.



- In the layers panel, click on the text layer.
- Right Click and select **Blending Options**
- Add some blending options. Play around with them to see if you can make your text really stand out!

Experimental!

- Your task for the rest of the lesson is to experiment with Photoshop to create a Simpsons scene with different characters (each character should have its own layer).

- You should try to use different layers, text and tools to create your scene.
- You have a few options of backgrounds:







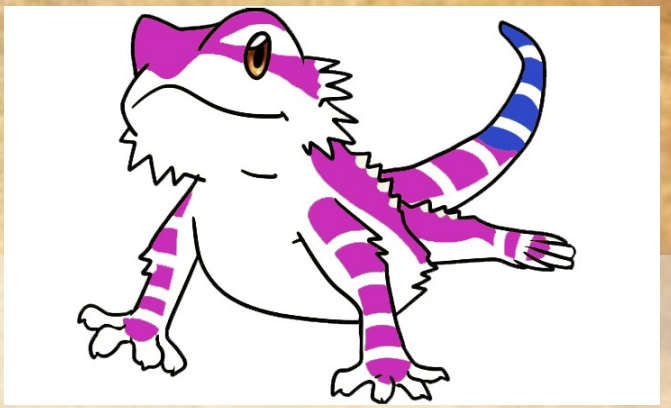


Checkpoint!

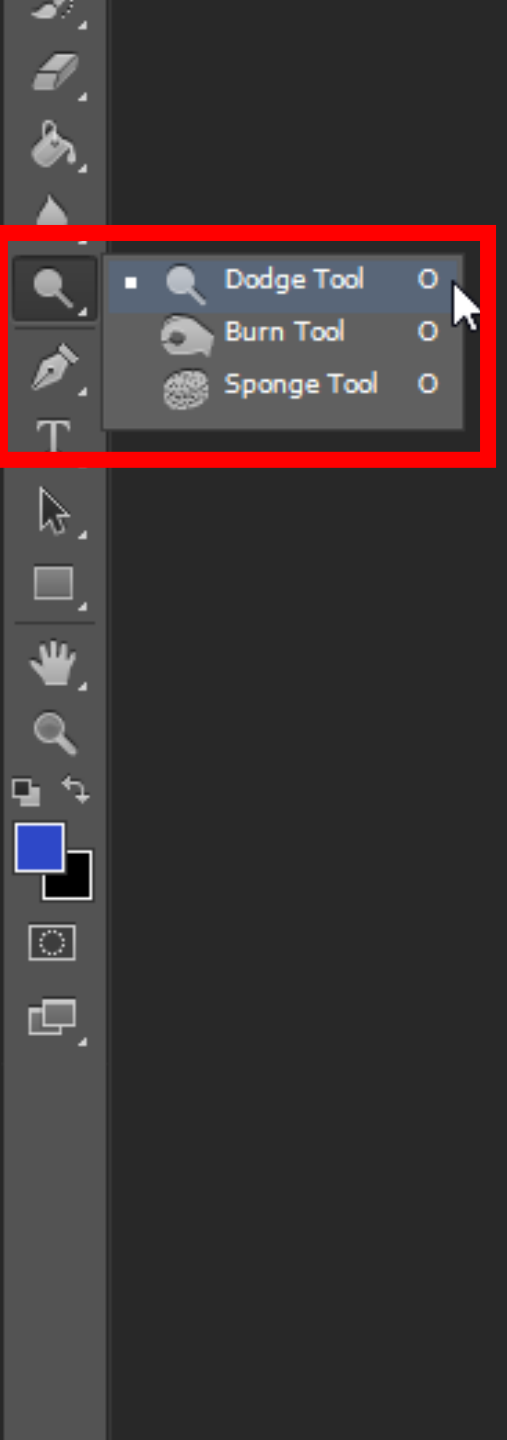


- So far, we have been able to:
  - Insert an image, placing them into layers
  - Resize layers
  - Clone parts of an image
  - Remove background from an image
  - Add text onto a graphic
  - Add blending options
  - Fill in areas of an image with colour
  - Save Photoshop files
  - Export Photoshop files as a png image

The Next Steps...



- We will now be looking to make a 2D image look 3D.
- Open Activity 3 in Photoshop.
- Using the Paint Bucket Tool, you need to colour in your dragon first of all.
- Remove the Background from your image.
- When Done, turn off your monitor.



The tools we are now going to use are the **Burn** and **Dodge** tools.

Q. Can anyone remember what these do?

Burn Tool

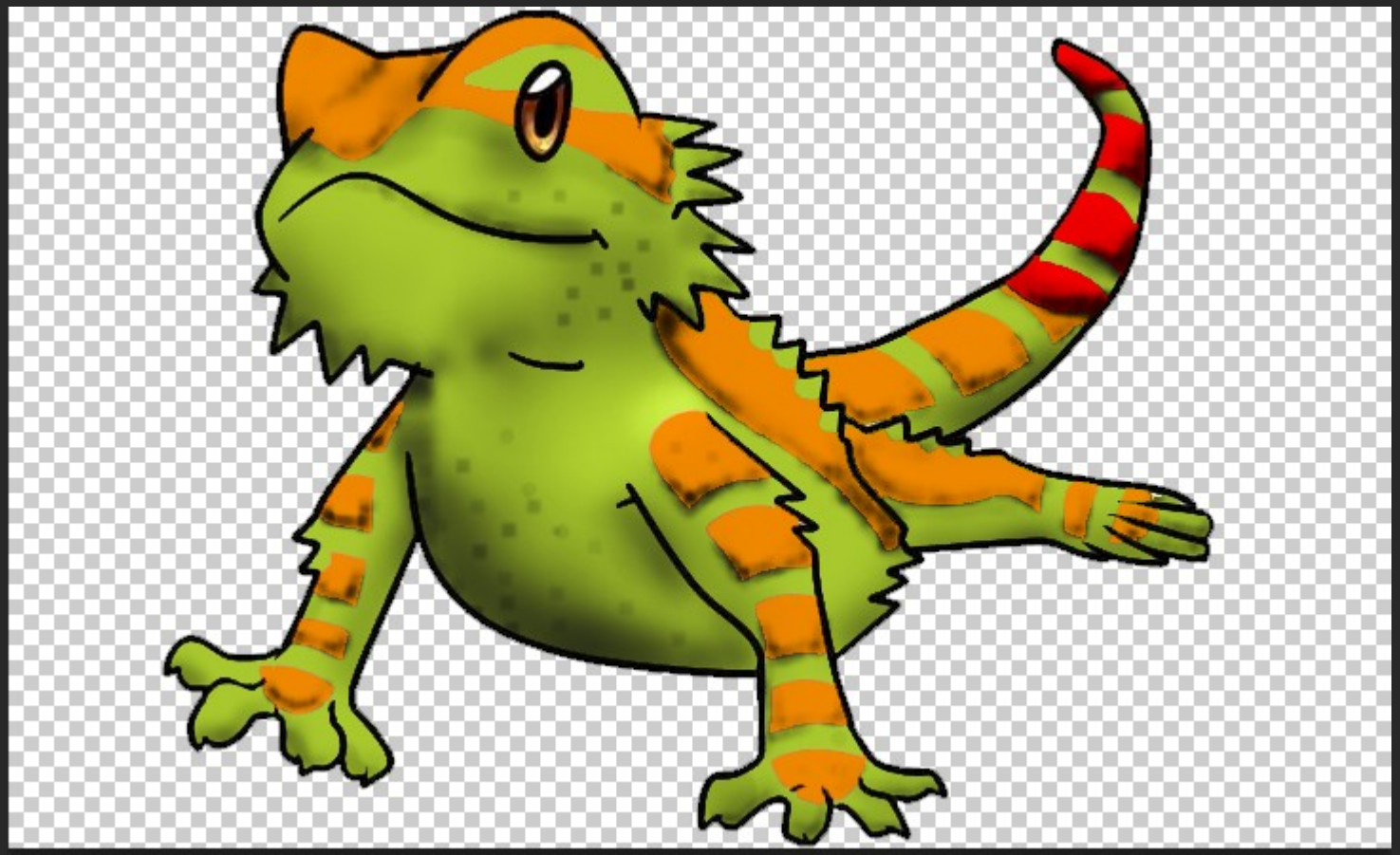
Uses a brush to **darken** parts of an image

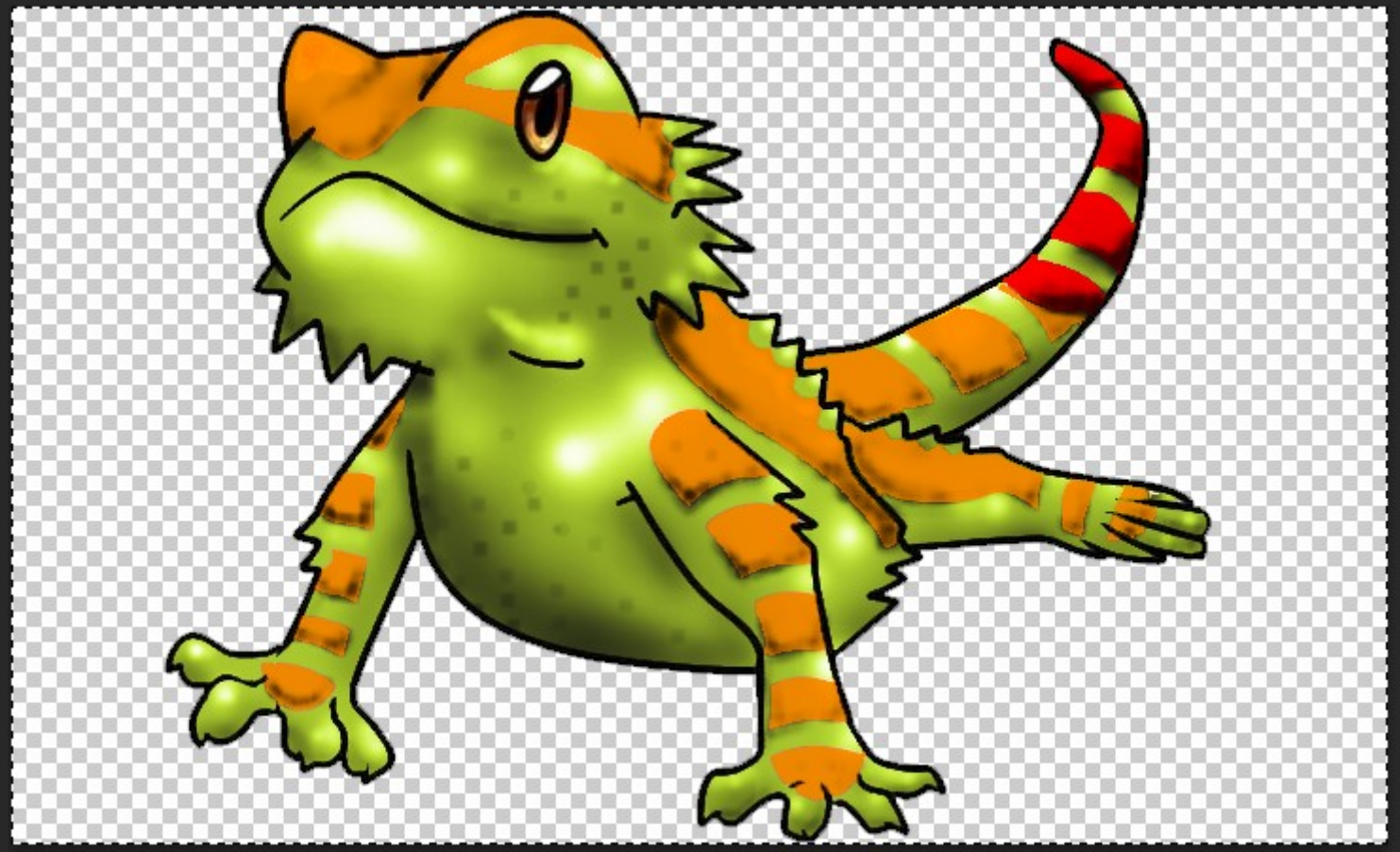
Dodge Tool

Uses a brush to **lighten** parts of an image

Amazing Hint!
Use [and] on your keyboard to change the brush size!

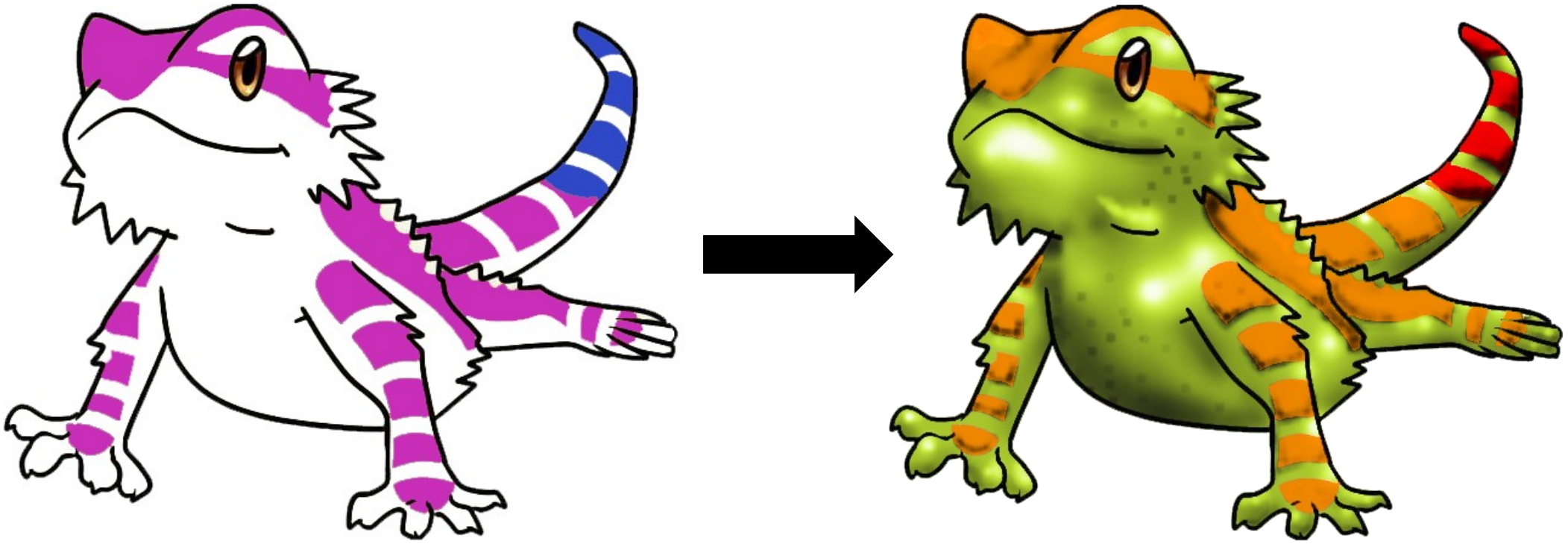
You have 5 minutes now to use the Burn Tool to add shadowing.





You have 5 minutes not to use the Dodge Tool to add shine + highlights.

Export your Photoshop file as a PNG Image.



Layering Images

- We are now going to be working with layers in Photoshop.
- This is a bit like sticking things down onto a piece of paper – we can create some pretty cool and unique graphics by using layers!
- Our next exercise is to add layers to images.

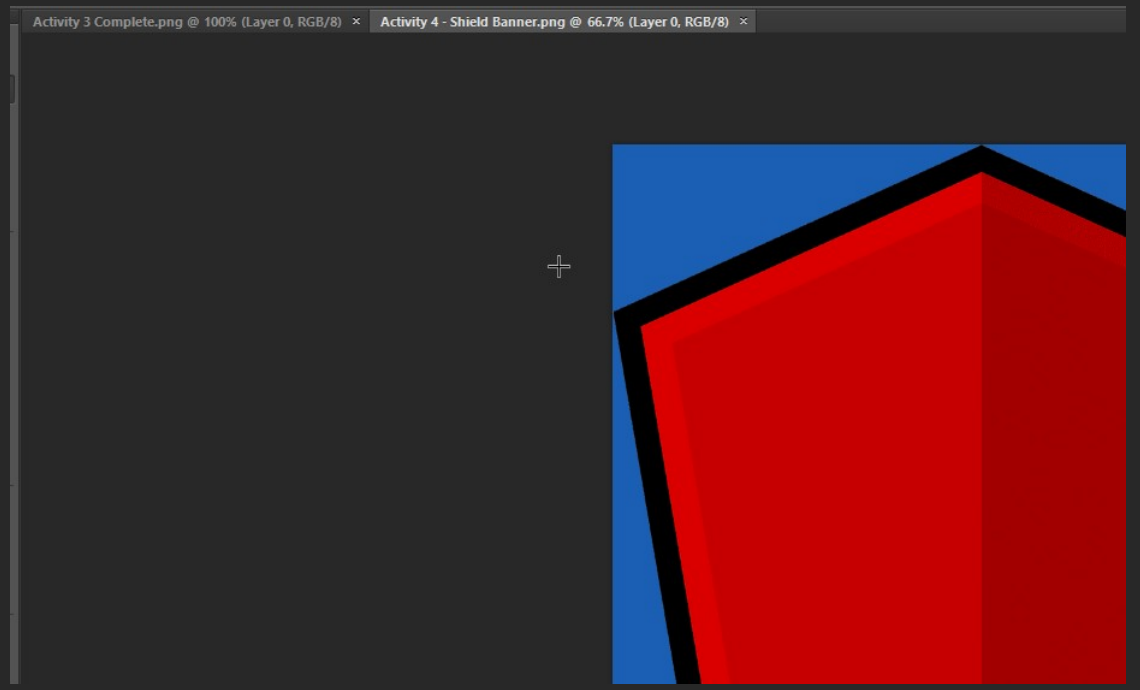


Adobe Photoshop

STEP: 13

Tool/Skill Used: Photoshop Layers

Purpose: Allows us to layer our images on top of each other



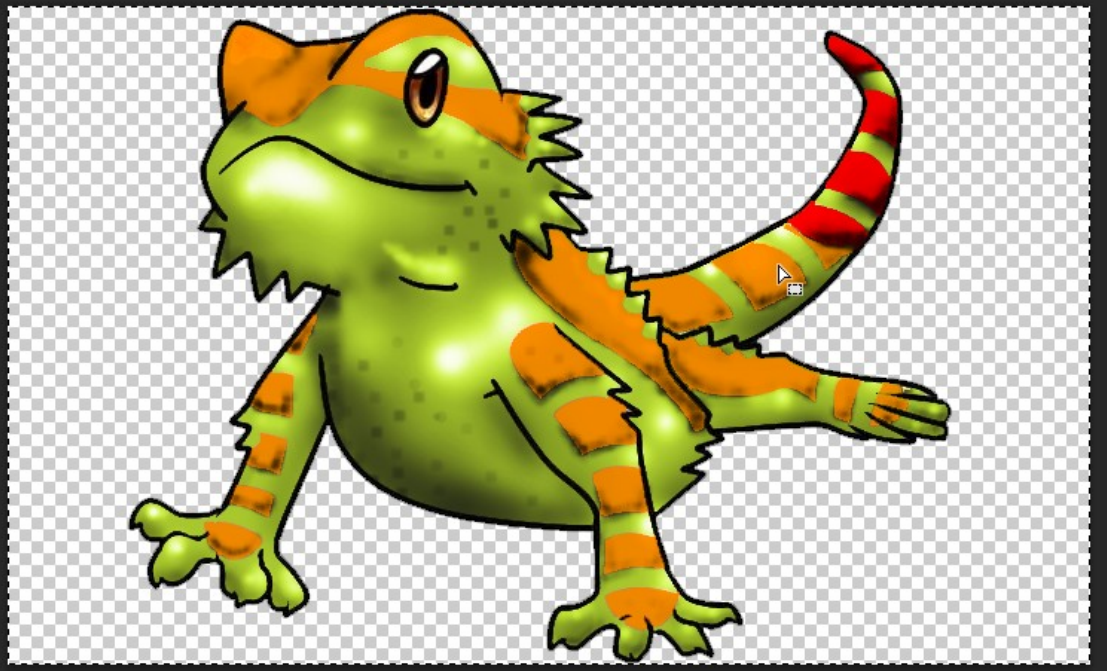
- Save ‘Activity 4 – Shield Banner’ into your area under your Photoshop folder.
- Open this in Photoshop
- Also Open the image of your Lizard/dragon from earlier.

Adobe Photoshop

STEP: 14

Tool/Skill Used: Copy and Pasting!

Purpose: Place two files into one image



- Click on your Activity 3 image.
- Press **Ctrl** + **A** to select all. You will get a matching border around the whole image.
- Press **Ctrl** + **C** to Copy this selection.

Adobe Photoshop

STEP:	15
Tool/Skill Used:	Pasting!
Purpose:	Place two files into one image

- Click on your Activity 4 tab.
- Press **Ctrl** + **P** to paste your image into a new layer above your shield banner.



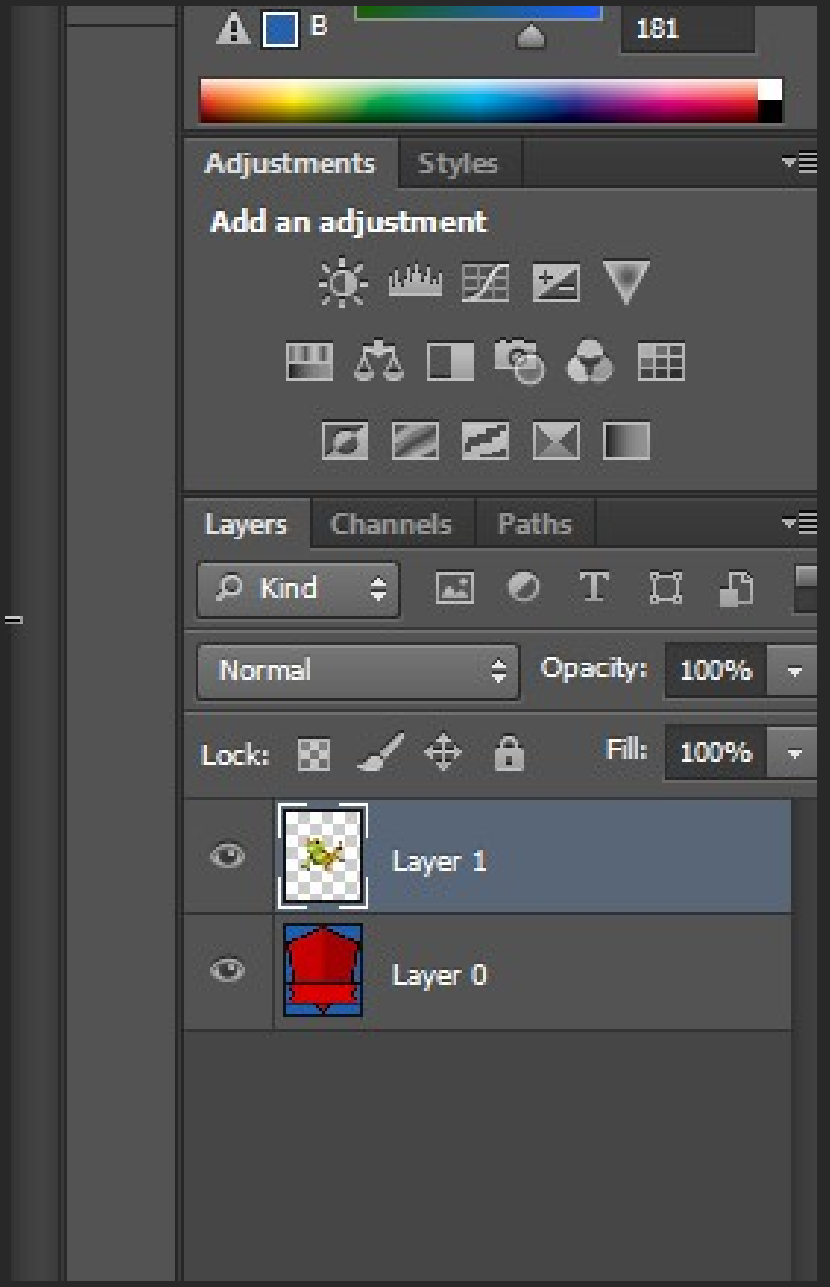
Adobe Photoshop

STEP: 15

Tool/Skill Used: Layers

Purpose: Selecting Layers

- The Lizard is on its own layer (Layer 1) with the Shield in Layer 0.
- We can move these layers independently from each other. So make sure you have clicked on Layer 1.



Adobe Photoshop

STEP: 15

Tool/Skill Used: Layers

Purpose: Moving Layers



- Using the Move Tool (V) in the toolbox, you can move the lizard further up the screen so it isn't covering the bottom part of the banner.
- Now add a text layer and put your name in the banner at the bottom. Position this also.



You have this lesson to change colours of the shield – make it your own! Add your own background instead of the blue (Note. You should do this by adding a new layer!).

Save your work as ‘Activity 4 – My Crest’

- We will be peer assessing this work before we continue to the assessed task.

- This will be in the form of ‘Two Stars and a Wish’

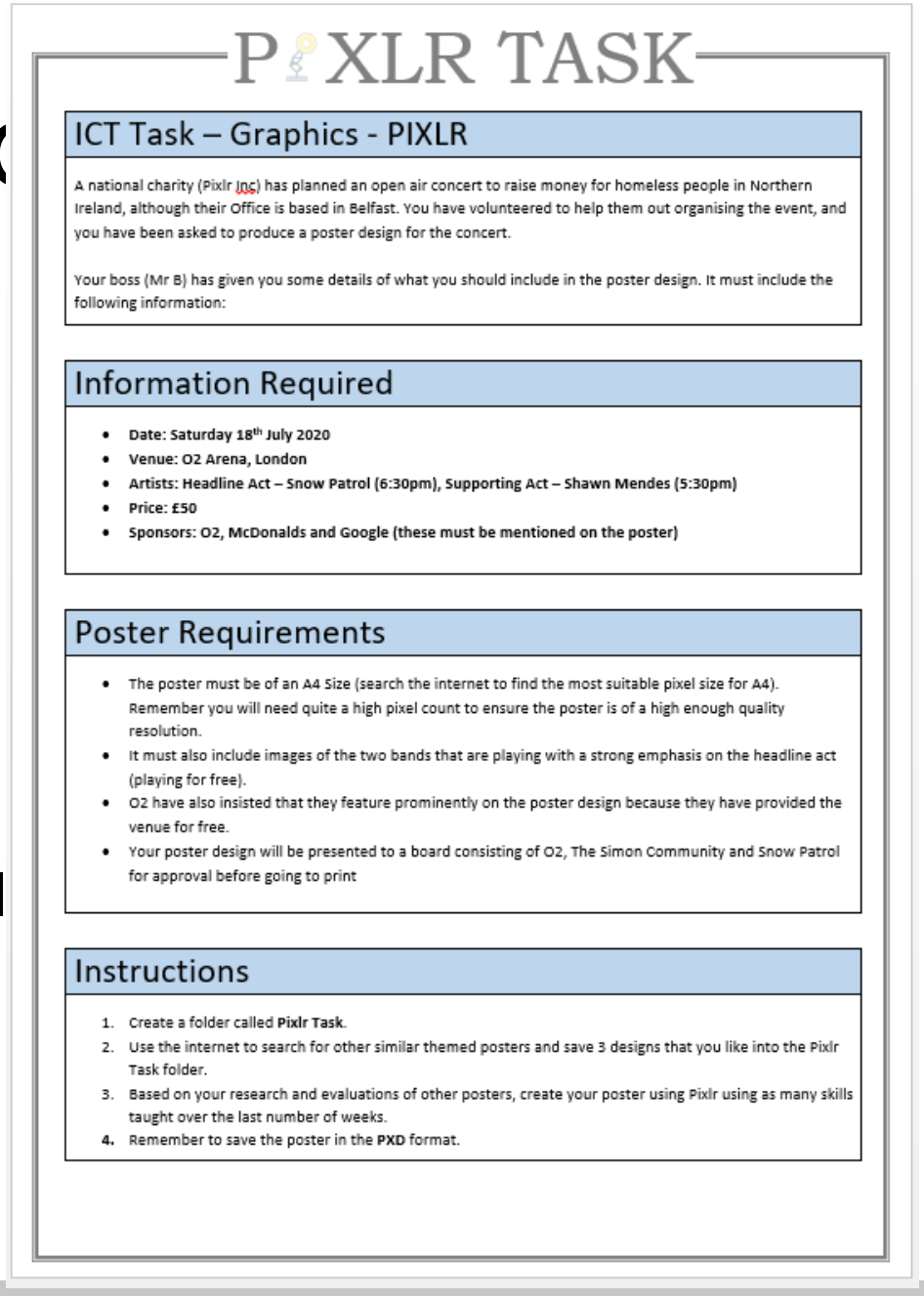
  - Two good things you really liked about your peers work
  - One improvement they could make to improve next time

- When you have finished, you will review your work by stating **one target you have before starting the next piece of work.** Take note of this in a Microsoft Word document called ‘Graphic Improvements’.

Assessed Task: Pixlr Inc

- You have a total of **three** lessons to complete 'ASSESSMENT - ICT PIXLR TASK' assessment.

- You will be marked on:
  - Research and its inspiration for your own poster
  - Creativity
  - Planning and Organisation (including Files and Fol
  - Use of Layers + other tools
  - Colour Scheme (3-5 Colours)
  - Including key information
  - Presentation and use of logos/brand names



PIXLR TASK

ICT Task – Graphics - PIXLR

A national charity (Pixlr Inc) has planned an open air concert to raise money for homeless people in Northern Ireland, although their Office is based in Belfast. You have volunteered to help them out organising the event, and you have been asked to produce a poster design for the concert.

Your boss (Mr B) has given you some details of what you should include in the poster design. It must include the following information:

Information Required

- **Date: Saturday 18th July 2020**
- **Venue: O2 Arena, London**
- **Artists: Headline Act – Snow Patrol (6:30pm), Supporting Act – Shawn Mendes (5:30pm)**
- **Price: £50**
- **Sponsors: O2, McDonalds and Google (these must be mentioned on the poster)**

Poster Requirements

- The poster must be of an A4 Size (search the internet to find the most suitable pixel size for A4). Remember you will need quite a high pixel count to ensure the poster is of a high enough quality resolution.
- It must also include images of the two bands that are playing with a strong emphasis on the headline act (playing for free).
- O2 have also insisted that they feature prominently on the poster design because they have provided the venue for free.
- Your poster design will be presented to a board consisting of O2, The Simon Community and Snow Patrol for approval before going to print

Instructions

1. Create a folder called **Pixlr Task**.
2. Use the internet to search for other similar themed posters and save 3 designs that you like into the Pixlr Task folder.
3. Based on your research and evaluations of other posters, create your poster using Pixlr using as many skills taught over the last number of weeks.
4. Remember to save the poster in the **PXD** format.